

Report 10 - Literature Model Families Source Catalogue

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Abstract

This report records how the literature model families identified in Report 9 have been mapped into the OpenHPLjl source tree. It distinguishes baseline models, runnable prototypes, and planned high-fidelity families that require additional interfaces, data, or spatial discretization.

1 Checkpoint

The previous checkpoint established the core chain

Reservoir $\rightarrow$ RigidPipe $\rightarrow$ SurgeTank $\rightarrow$ Turbine $\rightarrow$ Shaft $\rightarrow$ Generator $\rightarrow$ Grid.

The current branch expands each folder into a literature-guided model family.

2 Status convention

- **baseline:** implemented in v0.1.0-alpha.1.
- **prototype:** source model or executable structural prototype added.
- **planned:** model family documented, but production implementation requires new interfaces, calibration data, or PDE/discretization machinery.

3 Source catalogue

Domain	Model family	Status	Source direction
Interfaces	HydraulicPort; RotationalPort; ElectricalPowerPort	baseline	Current acausal interfaces.
	PhasorPort	prototype	Real/imaginary voltage-current connector for machine/network DAEs.
Reservoirs	InfiniteReservoir; Reservoir	baseline	Constant head and constant-area storage.
	NonlinearReservoir	prototype	Area-elevation relation $A(H)$.
	ReservoirChannel	planned	Reservoir plus Saint-Venant open-channel equations.
Friction	Quadratic, Darcy, laminar, Haaland, Swamee-Jain	baseline	Current friction registry.
	Colebrook-White residual; transitional blend	prototype	Implicit turbulent and blended regime models.
	Unsteady friction	planned	Frequency-dependent transient friction.
Waterways	QuasiSteadyPipe	prototype	Algebraic head-loss conduit.
	RigidPipe	baseline	Rigid water-column ODE.
	ElasticPenstock	prototype	Elastic PDE family and intended semi-discrete structure.
	MOCWaterway; FiniteVolumeWaterway	planned	Distributed water-hammer solvers.
Surge tanks	SurgeTank	baseline	Simple open tank.
	Throttled; Orifice; AirCushion; VariableArea	prototype	Alternative storage and restriction/gas-law families.
	DifferentialSurgeTank	planned	Multi-chamber surge system.

Domain	Model family	Status	Source direction
Turbines	IdealTurbine; SimpleGateTurbine	baseline	Reduced conversion and gate-flow models.
	TurbineLookup; Francis; Pelton; Kaplan	prototype	Hill-chart/data model and turbine-type-specific families.
Shafts	LumpedShaft	baseline	Single inertia.
	TwoMassShaft	prototype	Two inertias with torsional spring-damper.
Generators	MultiMassShaft	planned	N-mass torsional chain.
	IdealGenerator	baseline	Reduced mechanical-electrical converter.
	ClassicalGenerator; FourthOrderGenerator	prototype	Swing and transient-emf families.
Grids	SixthOrder; SalientPole; RoundRotor	planned	Subtransient and machine-type-specific models.
	InfiniteGrid	baseline	Fixed-frequency boundary.
	AggregateGrid; SMIBGrid	prototype	Finite-inertia aggregate and power-angle network families.
	MultiMachineGrid; EMTGrid	planned	Phasor-network and electromagnetic-transient end-points.

4 Registry

The branch adds `src/ModelFamilies/ModelFamilyRegistry.jl`. The registry provides one machine-readable catalogue grouped by physical domain and tagged with `:baseline`, `:prototype`, or `:planned`.

This is important because future tutorials, tests, and documentation can query the same catalogue instead of duplicating model lists.

5 Models wired into the package

The following new families are sufficiently compatible with the current connectors to be included directly:

- PhasorPort,
- NonlinearReservoir,
- transitional friction factor,
- QuasiSteadyPipe,
- TwoMassShaft,
- model-family registry.

The higher-fidelity turbine, generator, grid, elastic-waterway, and specialized surge-tank files are intentionally kept as prototypes/specifications until their required data structures and interfaces are finalized.

6 Validation rule

A literature family is not promoted from prototype to baseline merely because a file exists. Promotion requires:

1. complete governing equations,
2. parameter and unit table,
3. structural equation/unknown check,
4. analytical or limiting-case test,
5. component simulation,
6. connected-system test,
7. comparison with literature/reference behaviour.

7 Next implementation wave

The preferred promotion sequence is:

TurbineLookup → ClassicalGenerator → SMIBGrid → TwoMassShaft → ThrottledSurgeTank → ElasticPenstock.

Conclusion

All literature families identified in Report 9 now have an explicit location in the OpenHPLjl source architecture. The next work is no longer catalogue creation; it is model-by-model promotion from specification to validated ModelingToolkit implementation.